

Signals & Systems (EE2008)

Final Exam

Date: May 21, 2025

Course Instructor(s)

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Total Time (Hrs): 3

Total Marks: 100

Total Questions: 6

Roll No

Section

Student Signature

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Attempt all the questions.

CLO # 1: Express different types of signals and systems.

Q1: Express the types of operations involved in the given signal $x(t)$ as shown in Figure 1 for the following transformations. Also, sketch the resulting signals derived from $x(t)$. [5+5+5+5 marks]

a) $x(t-3)$

b) $x(t/2)$

c) $x(-t)$

d) $x(2t-4)$

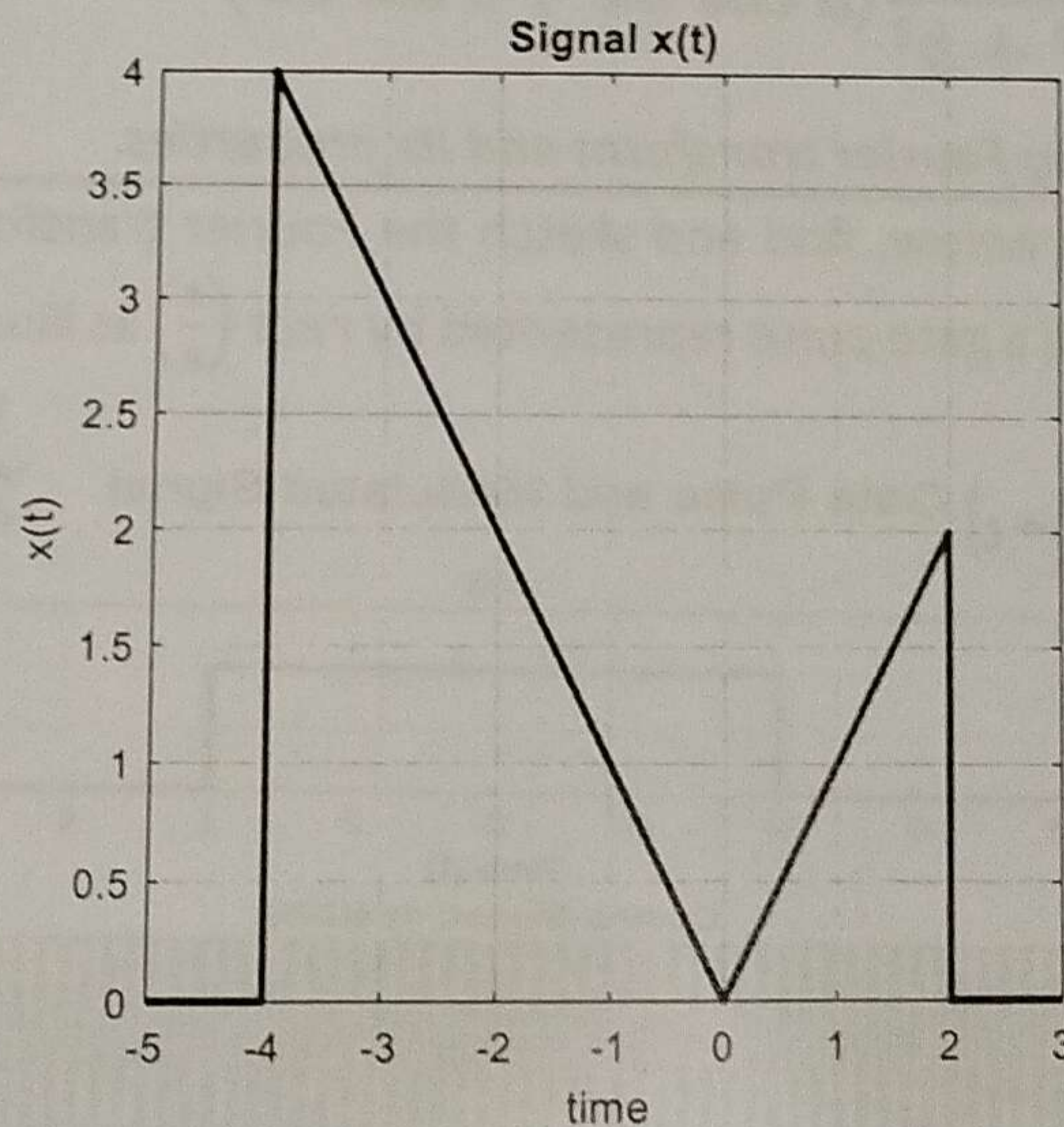


Fig-1

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\frac{y-4}{x+4} = \frac{-4}{4}$$

$$\frac{y-4}{x+4} = -1$$

$$y-4 = -4-x$$

$$y = -x$$

$$\frac{y-4}{x+1} = \frac{-4}{3}$$

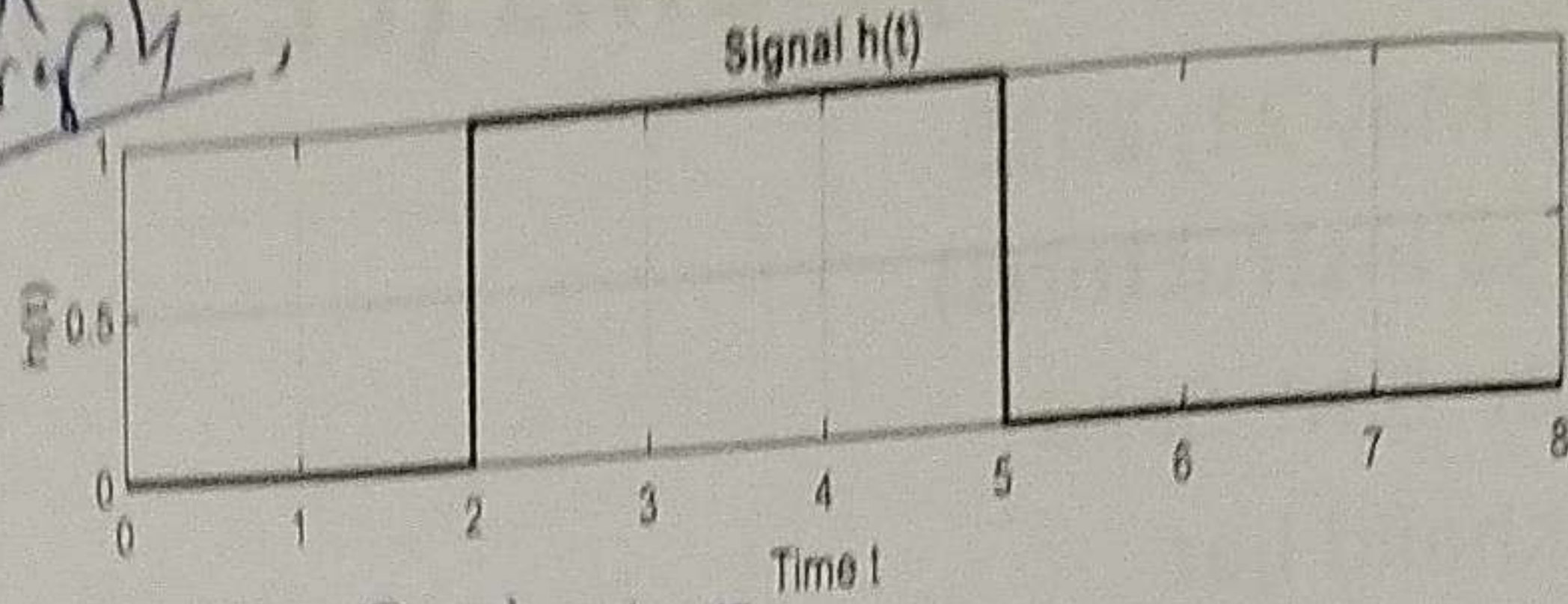
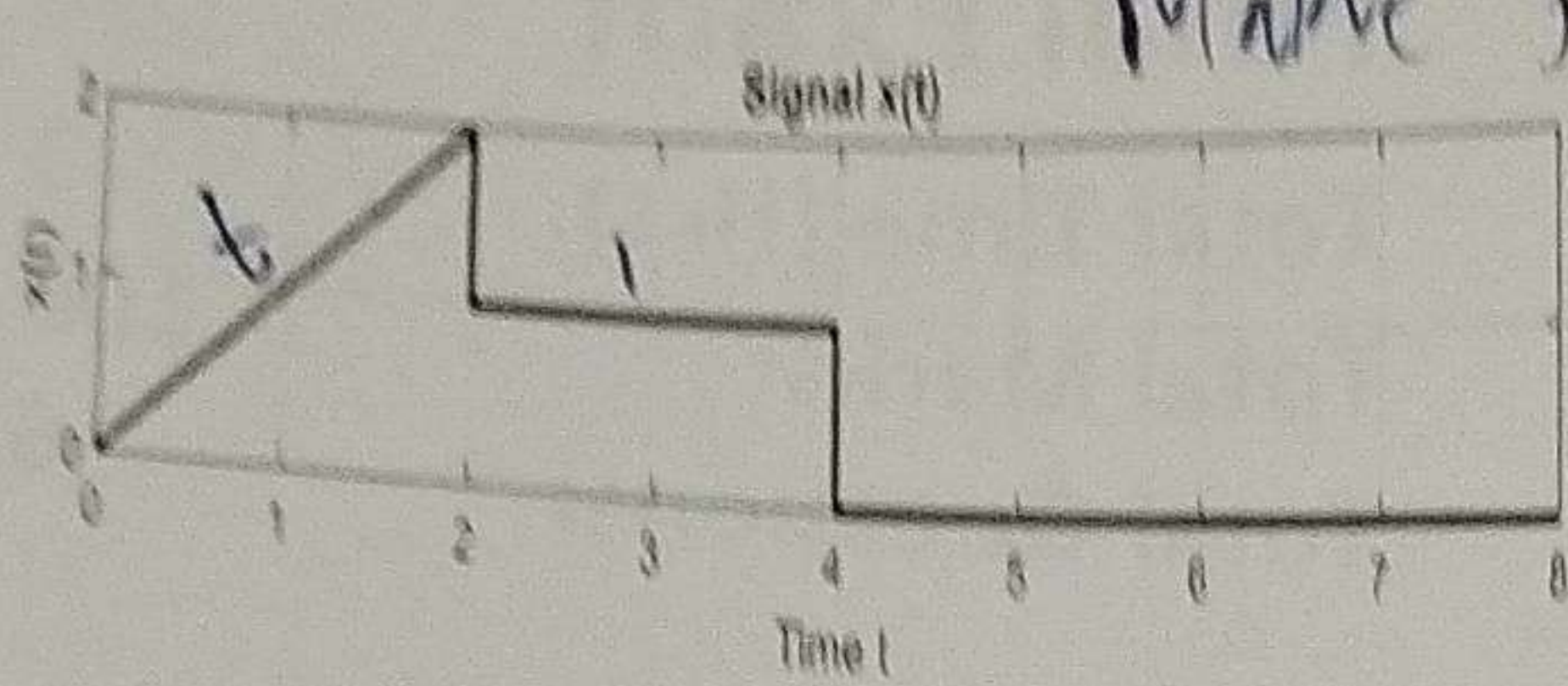
$$y-4 = -4x-4$$

$$y = -4x$$

$$y = -4t$$

CLO # 2: Apply mathematical and graphical convolution techniques to analyze CT and DT systems.

Q2: Consider the following two signals $x(t)$ and $h(t)$ as shown in Figure 2. Apply graphical convolutional technique to compute the convolution of these two signals. [20 marks]



Make Simul graph

Fig-2

CLO # 3: Analyze signals using Fourier series representation.

Q3: Analyze the periodic signal shown in Fig-3, calculate all the coefficients of trigonometric Fourier series and also write the final expression of trigonometric Fourier series. [20 marks]

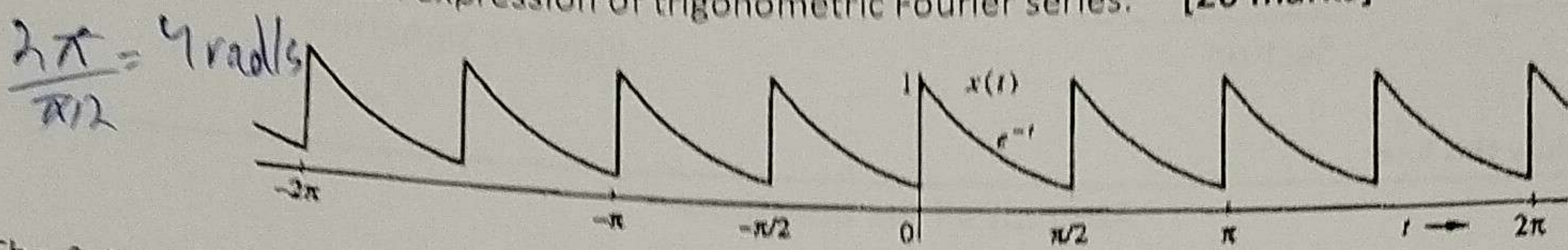


Fig-3

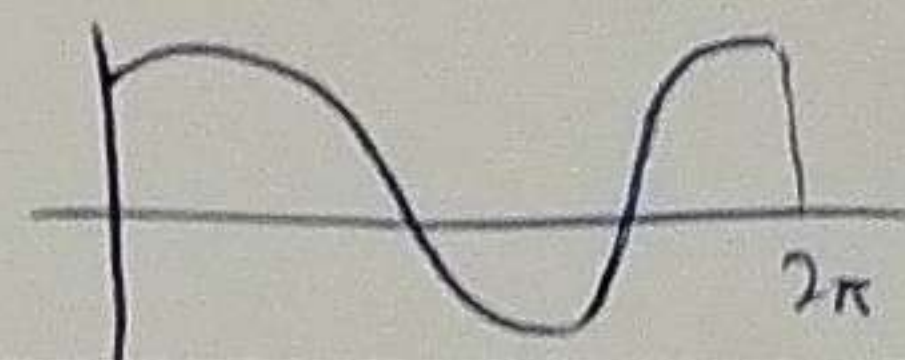
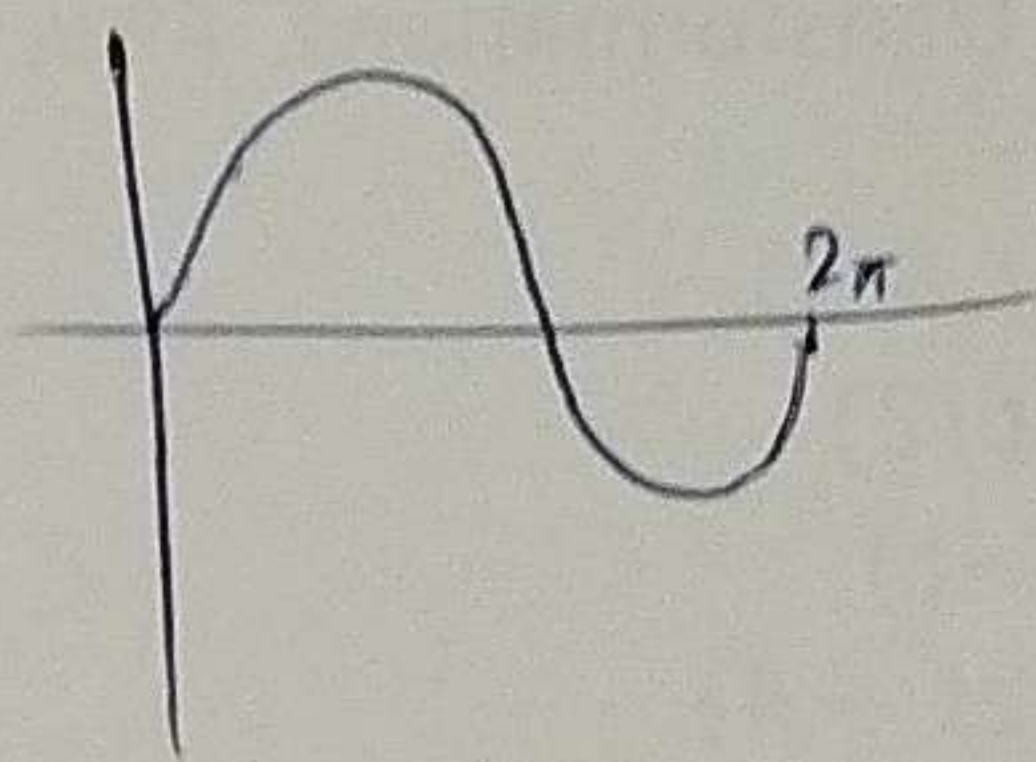
The following mathematical expressions can be used:

$$\int x^2 e^{ax} dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2)$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

$$x(t) = C_0 + \sum (C_n \cos n\omega_0 t + D_n \sin n\omega_0 t)$$



CLO # 4: Analyze signals using Fourier transform and its properties.

Q4: Analyze the signal given below, find and sketch the Fourier transform of the modulated signal $x(t) \cos(20t)$ in which $x(t)$ is a gate pulse represented by $\text{rect}(\frac{t}{8})$ as illustrated in figure given below. [15 marks]

Gate Pulse and Modulated Signal

$$\frac{1}{2} (X(\omega + 20) + X(\omega - 20))$$

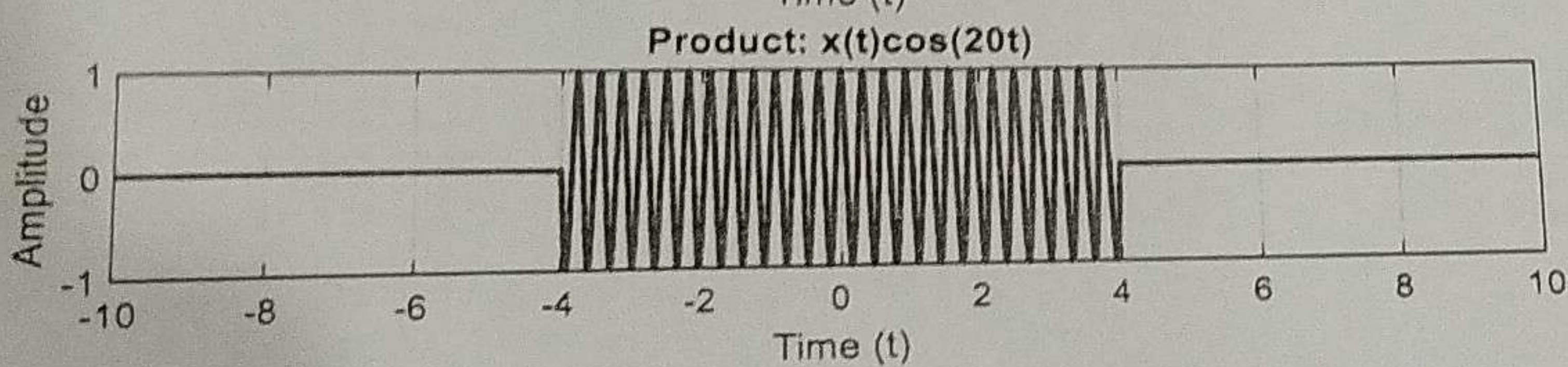
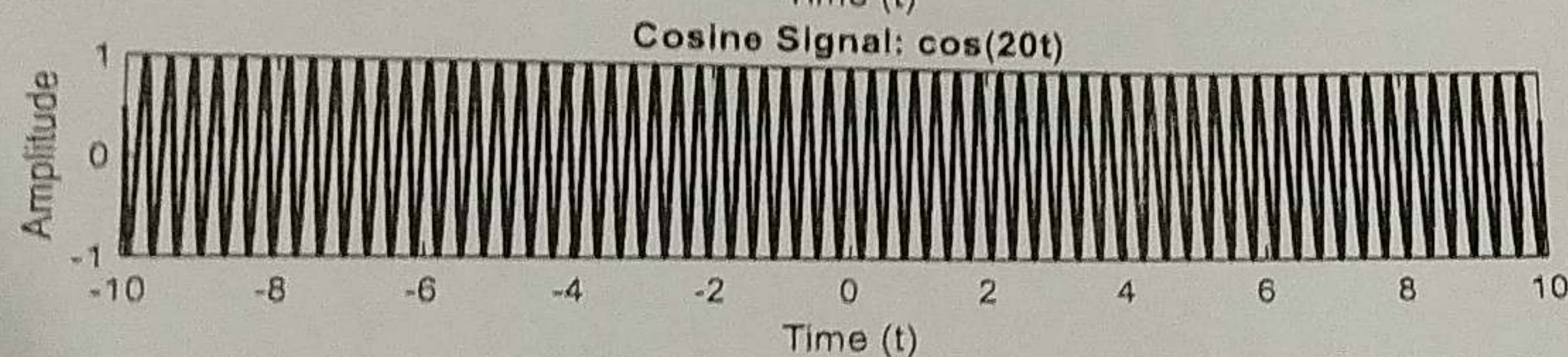
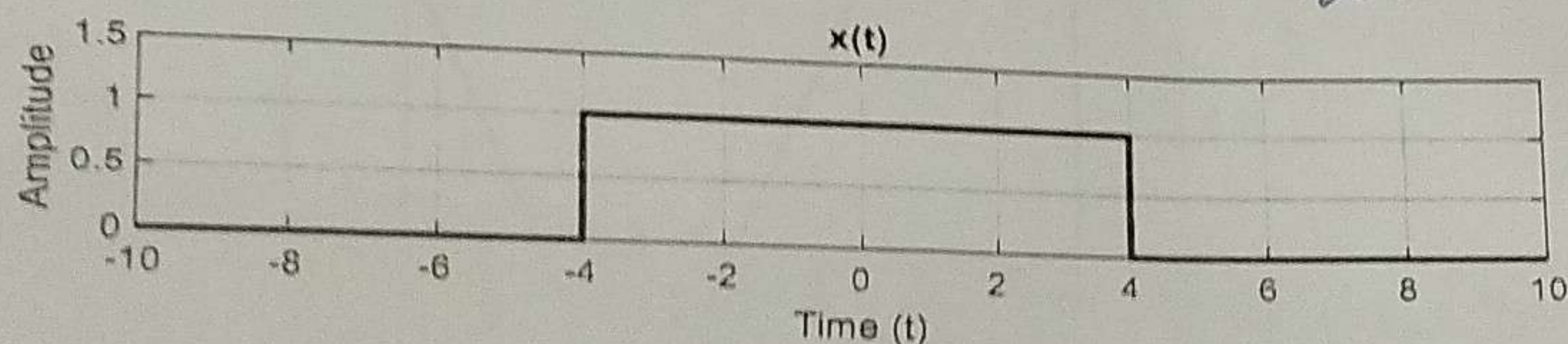


Fig-4

$$\begin{aligned} & 2 \sin \omega t \\ & 8 \sin \frac{\omega t}{2} \end{aligned}$$

$$\text{rect}\left(\frac{t}{8}\right) = 2 \text{sinc}\left(\frac{\omega t}{2}\right)$$

CLO # 4: Analyze signals using Fourier transform and its properties.

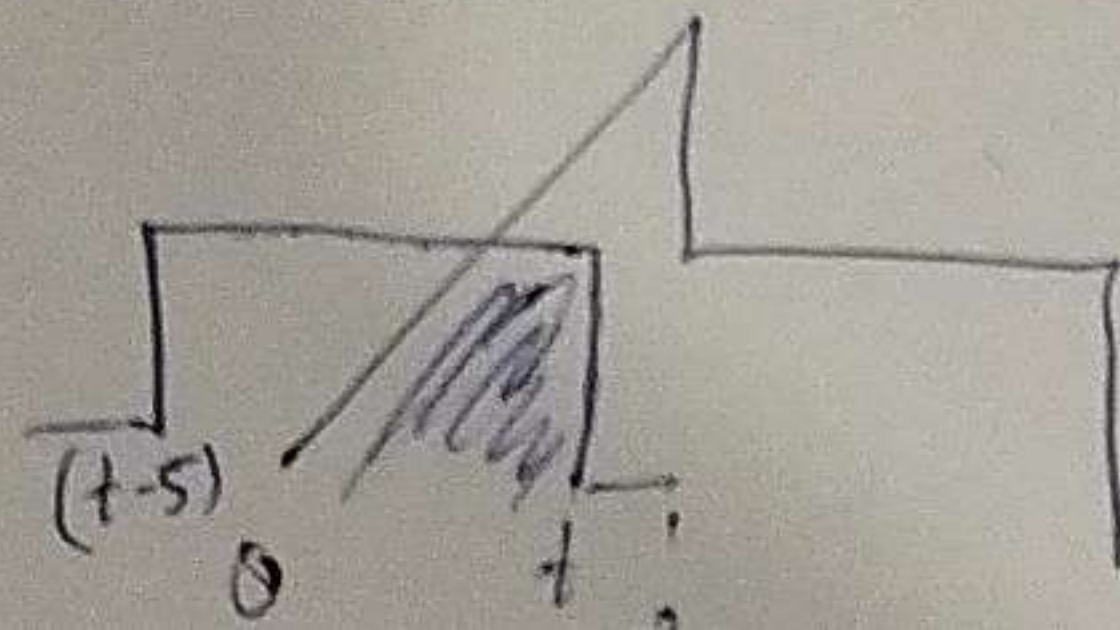
Q5: Analyze and find the Fourier transform of $e^{-|t|} \cos(2t)$ using appropriate property of Fourier transform. [15 marks]

CLO # 5: Describe and use the sampling theorem.

Q6: A television signal (video and audio) has a bandwidth of 5 MHz. This signal is sampled, quantized, and binary coded to obtain a PCM signal. [10 marks]

- Describe** what the sampling rate will be if the signal is to be sampled at a rate 25% above the Nyquist rate.
- If the Nyquist samples are quantized into 1024 levels and then binary coded, **describe** the appropriate number of binary digits required to encode a sample. $\rightarrow \log_2 L =$
- Describe** the appropriate number of binary pulse rate (bit/s) required to encode the video and audio signal.

$f_s \times \text{pulse rate binary}$



$$\frac{1}{2} (e^{j\omega t} + e^{-j\omega t})$$